GNSS L1 + L2/L5 Active Antenna Front End

RF circuit models, Touchstone files, ANSYS Circuit automation and simulation report

RFMW-ANT project 16 | 2026-11-01 | VDD +5 V | data: 8 supplied datasheets

1 Executive summary

This report models a dual-branch GNSS active-antenna front end built only from the eight documents supplied (Qorvo TQP3M9037, Taoglas HC125A and DXP.02.A, Qualcomm RF360 B7504 and B8889, the RF360 GNSS SAW product brief, the RF360 automotive SAW selection table and RF360 application note #18). The topology was confirmed by the user: two patch feeds per band are combined in an HC125A hybrid, followed by LNA - SAW - LNA - SAW, and the two bands are combined in the DXP.02.A SAW diplexer onto one RF output. All four LNAs run at +5 V / 70 mA.

Band	f (MHz)	Gain (dB)	Ripple (dB)	NF (dB)	In VSWR	Out VSWR	GD (ns)	GD ripple (ns)
L1/E1/B1C	1575.42	38.60	0.34	0.74	2.29	1.47	38.0	0.1
B1I	1561.10	37.69	0.51	0.74	1.49	1.35	44.4	6.2
G1	1602.00	37.56	0.14	0.73	2.07	1.26	41.8	9.7
L5/E5a/B2a	1176.45	41.99	1.14	0.59	1.63	1.42	27.3	14.9
E5b/B2b	1207.14	41.55	0.42	0.60	2.75	2.17	24.2	5.2
L2	1227.60	40.14	0.41	0.62	1.96	1.58	32.0	0.8
G2	1248.00	33.40	12.37	0.67	1.52	1.36	80.1	33.1

Table 1 - Simulated system performance at the band centres (virtual RHCP input to RF output, 50 ohm, 25 C). Ripple and group-delay ripple over the signal bandwidth. Group delay is an estimate (min-phase SAW models).

Key findings:

- Noise figure 0.74 dB at L1 and 0.59 dB at L5, set almost entirely by the hybrid loss (0.36 / 0.19 dB) plus LNA 1. The second LNA, both SAWs and the 3.1-3.6 dB diplexer loss add < 0.02 dB.
- Gain 38.6 dB (L1) and 42.0 dB (L5) - higher than most GNSS receivers want (typ. 15-30 dB at the receiver input after the cable). See recommendations.
- All blocks and both branches are unconditionally stable: LNA K >= 1.014, mu >= 1.057 over 0.05-5 GHz; branch mu >= 1.035.
- Input match at the RHCP port is only VSWR 2.29 at L1 and 2.75 at E5b: the LNA S11 (-10 dB) reflects through the hybrid back to the antenna.
- GLONASS G2 (1246-1256 MHz) is not usable: the B8889 pass-band ends at ~1245 MHz (gain ripple 12.4 dB across G2).
- Interference: an aviation DME/TACAN pulse of -30 dBm at 1150 MHz is attenuated only ~4 dB by each B8889 and reaches +13.3 dBm at LNA 2 output - above the B8889 input rating (+5 dBm) and only 8.3 dB below LNA 2 P1dB; Iridium (1621 MHz, -20 dBm) reaches +12.0 dBm in the L1 branch. There is no filter in front of LNA 1, so cellular uplinks at -10 dBm leave only ~7-10 dB back-off at LNA 1.
- Four independent circuit solvers agree: scikit-rf (reference), heavi, circuitRF and Qucsator-RF differ by < 0.01 dB in band. The ANSYS AEDT run is described in section 10.

Data quality: only the LNA S-parameters and noise parameters, the HC125A typical table and the datasheet spec tables are measured/extracted values. The SAW and diplexer Touchstone files are ESTIMATES built from digitized curves with a minimum-phase reconstruction; they reproduce every datasheet insertion-loss and attenuation table entry (section 11) but are not measured data. Replace them with vendor .s2p/.s3p files when available.

2 System architecture and connection diagram

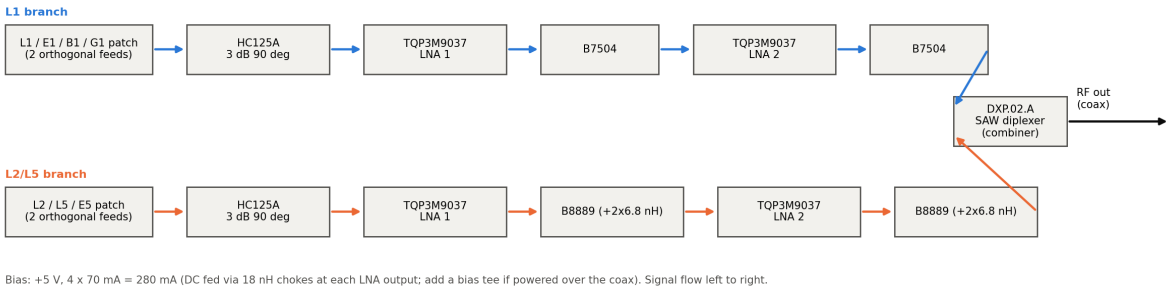


Figure 1 - Connection diagram (user-confirmed topology).

Signal path per branch: two orthogonal patch feeds -> HC125A 3 dB quadrature hybrid (pins 3/4 in, pin 1 out, pin 2 terminated in 50 ohm) -> TQP3M9037 LNA 1 -> SAW -> TQP3M9037 LNA 2 -> SAW -> DXP.02.A port (L1: pin 7, L2/L5: pin 5). The DXP.02.A is used in reverse as a combiner (reciprocal SAW device) and drives the common RF output (pin 2).

Ref	Part	Manufacturer	Function	Qty	Data used
U1-U4	TQP3M9037	Qorvo	LNA 0.7-6 GHz, NF 0.4 dB, 20 dB	4	S-param + noise tables (5 V)
HY1, HY2	HC125A	Taoglas	3 dB 90 deg hybrid 1150-1630 MHz	2	Typical table p.6
F1, F2	B7504 (B39162B7504L210)	Qualcomm RF360	SAW L1/B1/G1 1559-1606 MHz	2	Spec table + digitized plots
F3, F4	B8889 (B39122B8889P810)	Qualcomm RF360	SAW L5/E5/B2/L2 1166-1229 MHz (+2x6.8 nH)	2	Spec table + digitized plots
DX1	DXP.02.A	Taoglas	SAW diplexer L1 / L2+L5	1	Spec table + digitized plots
-	SAW product brief, SDST, AN #18	Qualcomm RF360	Alternatives / matching method	-	Reference only

Table 2 - Component summary.

3 Extracted datasheet parameters

All extracted numbers are in Scripts/ds_data.py (single source) and exported to Extracted_Data/ as 43 CSV tables plus Extracted_Parameters.xlsx (one sheet per table, with an INDEX sheet naming source page and kind: TABLE = typed from a datasheet table, DIGITIZED = read off a plot, [ASSUMED] = estimate).

Parameter	Value	Source
TQP3M9037 gain / NF / OP1dB / OIP3 @1.95 GHz, 5 V	20.0 dB / 0.40 dB / +20 dBm / +35 dBm	Qorvo p.2
TQP3M9037 IDD / VDD range / abs. max RF in	70 mA / 3.0-5.25 V / +22 dBm	Qorvo p.2
TQP3M9037 eval @1.7 GHz (5 V)	21.4 dB, NF 0.36 dB, OP1dB 20.7, OIP3 34.9 dBm	Qorvo p.6
TQP3M9037 noise parameters	1.5-2.7 GHz: NFmin 0.29-0.51 dB, |Gopt| 0.02-0.16, Rn 0.04-0.08 ohm	Qorvo p.3
HC125A IL / isolation / VSWR / balance	<= 0.3 dB / >= 22 dB / 1.2 / +/-0.35 dB, 90 deg	Taoglas p.5
HC125A typical @1582 MHz	C -3.25, T -3.27 dB, iso 28.6 dB, phase 92.97 deg	Taoglas p.6
B7504 IL typ/max	1.1/2.0 (B1), 0.9/1.4 (L1), 1.2/2.0 (G1) dB	RF360 p.7
B7504 min attenuation	45-60 dB cellular, 35-42 dB at 4.4-5.85 GHz	RF360 p.7
B7504 max input power	+23 dBm @915 (GSM 1:8), +15 dBm CW @1463/1710 MHz	RF360 p.8
B8889 IL typ/max	0.9/1.5 (L5), 1.0/1.3 (B2b), 1.4/2.6 (L2) dB	RF360 p.7
B8889 matching / max input	6.8 nH shunt both ports / +5 dBm CW in-band	RF360 p.6, p.8
DXP.02.A IL	3.8 typ / 4.6 max (L1), 4.2 / 5.2 (L2/L5) dB	Taoglas p.4
DXP.02.A isolation / input power	>= 22 dB (typ 31-35) / 10 dBm	Taoglas p.4

Table 3 - Main extracted parameters (full tables in Extracted_Data/).

4 Design assumptions

Every value not printed in a datasheet. The same list is in Extracted_Data/assumptions_log.md and in the XLSX.

ID	Item	Assumption	Basis / how to close
A1	Topology	Two branches (patch -> HC125A -> LNA -> SAW -> LNA -> SAW) combined by the DXP.02.A, as confirmed by the user 2026-10-31. Single chain is not possible: B7504 (L1) and B8889 (L2/L5) have disjoint pass-bands.	User decision

ID	Item	Assumption	Basis / how to close
A2	Antenna	Each patch is an ideal dual-feed RHCP source: lossless 90 deg splitter, 50 ohm feeds (PatchCP_ideal_RHCP_feed.s4p). LHCP port terminated at T0 (290 K) for noise.	Replace with measured/HFSS patch S-parameters
A3	LNA noise below 1.5 GHz	Noise table starts at 1.5 GHz. NFmin 0.38-0.40 dB, Gopt = 0.15/14.1 deg, Rn = 0.08 ohm used at 0.6-1.2 GHz (bounded by the 50-ohm eval NF of 0.40 dB at 0.9 GHz and 0.36 dB at 1.7 GHz).	Ask Qorvo for low-band noise parameters
A4	LNA bias	VDD = +5 V, 70 mA (user choice); S-parameters at device leads; eval-board bias network (100 pF DC blocks, 18 nH choke, 100 pF + 1 uF bypass).	Datasheet p.2, p.5
A5	Parasitics	100 pF 0402: ESL 0.25 nH, ESR 0.15 ohm; 18 nH 0603: Q = 40 at 1.4 GHz, 0.12 pF self-capacitance (SRF ~3.4 GHz); bypass ESL 0.4 nH.	Replace with vendor S-parameter models of the chosen parts
A6	LNA interpolation	200 MHz table interpolated with a cubic spline on dB magnitude and unwrapped phase. Linear interpolation changes in-band gain by up to 1.4 dB (heavi test) -> a real model uncertainty.	Download the Qorvo fine-step .s2p
A7	LNA large-signal	OIP3/OP1dB at L-band interpolated from eval-board values at 0.9 and 1.7 GHz (34.4-34.9 dBm, 22-20.7 dBm).	Datasheet p.6
A8	HC125A phase	Datasheet gives magnitudes and the coupled-vs-through phase difference (91.9-93.0 deg) only. Absolute through phase = -90 deg x f/1390 MHz; reflection and isolation phase offsets chosen to keep the 4-port passive (165 / 270 deg).	Measured .s4p from Taoglas
A9	HC125A port mapping	'Transmission' column = pin1->pin4 (direct, angle theta), 'Coupling' column = pin1->pin3 (theta-90), from the pin table p.7. Feeds on pins 3/4, LNA on pin 1, pin 2 terminated in 50 ohm.	Confirm with Taoglas
A10	HC125A passivity	The 92 deg phase difference with 0.15-0.29 dB loss is slightly non-passive with arbitrary reflection phases; after the phase choice the matrix is scaled by <= 0.11 dB at the worst frequency.	-
A11	HC125A out of band	Values held at the 1150/1630 MHz edge outside 1150-1630 MHz. Out-of-band rejection therefore comes from the SAWs/DXP only (conservative).	-
A12	SAW phase	No measured phase in the datasheets. S21 phase = minimum-phase (Hilbert / real-cepstrum) reconstruction of the digitized magnitude; S11/S22 phase = S21 phase + 90 deg (lossless symmetric reciprocal condition). Group delay is therefore an estimate; ripple agrees with the brief (B7504 GD ripple 4 ns typ).	Use RF360 measured .s2p (available on rffe.qualcomm.com)
A13	SAW reflection out of band	|S11| = 0.95 sqrt(1-|S21|^2) outside the plotted VSWR range (reflective SAW stop-band), blended over 4 MHz.	-
A14	B8889 matching	The B8889 model includes the required 6.8 nH shunt inductors (datasheet curves are measured with them).	Datasheet p.6
A15	DXP model	Digitized curves include the Taoglas eval-board matching (L2 4.3 nH, C1 1.5 pF, L1 4.7 nH, L3 3.3 nH, C2 1 pF). Reflection phase = dominant transmission phase + 90 deg.	Measured .s3p from Taoglas
A16	Digitizing accuracy	Pass-band plots +/-0.3 dB, wide-band attenuation plots +/-2 dB, VSWR +/-0.1.	-
A17	PCB	Interconnects ideal (zero length). 50 ohm lines on the real board add ~0.02-0.05 dB/cm loss and phase.	Add MSL models once the layout exists
A18	Interferer levels	Interferer powers at the antenna feed (-10 dBm cellular UL, -20 dBm Iridium/Inmarsat UL, -30 dBm DME) are example values for a co-located handset/terminal.	Set from the product's EMC environment
A19	Signal levels	Minimum received power (0 dBic antenna): GPS L1 C/A -128.5 dBm, L5 -127.9 dBm, L2C -130 dBm, GLONASS G1 -131 dBm (signal ICDs).	-
A21	S-band topology	S-band patch (ideal dual feed) -> C2327J5003AHF -> TQP3M9037 -> TA1442A -> TQP3M9037 -> TA1442A -> separate S-band output (no diplexer). Antenna only mentioned, modelled as ideal RHCP source.	User request 2026-10-31
A22	S-band LNA match	Qorvo 2300-2600 MHz reference design (2.7 nH shunt, 2.4 pF series, 5.6 nH choke); inductor Q 30-50 @2.5 GHz, ESL/ESR [ASSUMED].	Datasheet p.10
A23	C2327J5003AHF model	Magnitudes from excess IL, amplitude balance, isolation and input return loss (same RL assumed at all ports); coupled phase = direct + 90 deg - phase balance; reflection/isolation phases chosen for passivity over 2-3 GHz -> excess IL 0.43 dB vs 0.3 typ / 0.4 max (conservative).	Measured .s4p from TTM
A24	TA1442A model	Digitized VNA plots p.3-4 with the printed marker values used exactly; S11 = S22 shape (markers differ by < 0.01); min-phase phase.	Measured .s2p from Tai-Saw
A25	NavIC S signal	2492.028 MHz, SPS BW 2.046 MHz, RS BW 16.5 MHz, minimum received power -132.3 dBm (0 dBic) used as reference values.	Verify against the IRNSS SPS ICD
A20	Temperature	All data at 25 C typical. No temperature, part-to-part or aging spread is modelled.	-

Table 4 - Assumptions log.

5 Models and equations

5.1 Component models

- TQP3M9037: datasheet S-parameter table (0.05-5 GHz, device leads) -> TQP3M9037_5V.s2p with the measured noise block; the circuit adds the eval-board bias network (Figure 3).
- HC125A: 4-port from the typical table (magnitudes, coupled-through phase difference) -> HC125A.s4p.
- B7504 / B8889: |S21| digitized from Fig. 4 and anchored to the attenuation table, |S11|/|S22| from VSWR Fig. 5/6, phase by minimum-phase reconstruction -> B7504.s2p, B8889_matched.s2p (the B8889 file includes the 6.8 nH inductors).
- DXP.02.A: 3-port from the digitized p.6-8 curves (eval matching included) -> DXP02A.s3p.
- Patch antenna: ideal dual-feed RHCP source -> PatchCP_ideal_RHCP_feed.s4p (placeholder for the real patch).
- Equivalent circuits: LNA stage = series C(ESL, ESR) - 2-port table - shunt [L(Q) || Cpar] + bypass C(ESL) - series C; SAW static model = capacitive (DC block function, AN #18 Fig. 9).

5.2 Equations

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VSWR = (1 + |G|) / (1 - |G|), RL = -20 log10|G|, |G| = (VSWR - 1)/(VSWR + 1)

Min-phase: ln H(w) = ln|H(w)| + j arg H(w), arg H = -Hilbert{ln|H|} (real-cepstrum folding)

SAW reflection phase: arg S11 = arg S22 = arg S21 + 90 deg (S11 S21* + S21 S22* = 0, lossless
symmetric 2-port)

Transducer gain G_T = |S21|^2 (50 ohm source and load); group delay tau = -d(arg S21)/d(omega)

RHCP excitation of the hybrid: a3 = 1/sqrt2, a4 = -j/sqrt2 -> b1 = (S13 - j S14)/sqrt2

Rollett K = (1 - |S11|^2 - |S22|^2 + |D|^2) / (2|S12 S21|), D = S11 S22 - S12 S21

Edwards-Sinsky mu = (1 - |S11|^2) / (|S22 - D S11*| + |S12 S21|) (mu > 1: unconditionally stable)

Passive noise (Twiss): C_Y = 2kT0 Re{Y}; C_A = T C_Y T^H, T = [[0, B], [1, D]]

Active noise: C_A = 2kT0 [[Rn, (Fmin-1)/2 - Rn Yopt*], [(Fmin-1)/2 - Rn Yopt, Rn |Yopt|^2]]

Cascade: C_A = C_A1 + A1 C_A2 A1^H; F = 1 + z^H C_A z / (2kT0 Re Zs), z = [1, Zs*]^T

Friis: F = F1 + (F2 - 1)/G1 + ...; 1/IIP3 = sum (G_before,i / IIP3_i) (linear units)

Noise floor N = kT0 B F G = -174 dBm/Hz + 10 log10 B + NF + G; output noise of both branches adds
in the DXP: F_L1,sys = (F1 G1 + F2 G2)/G1
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6 Schematics

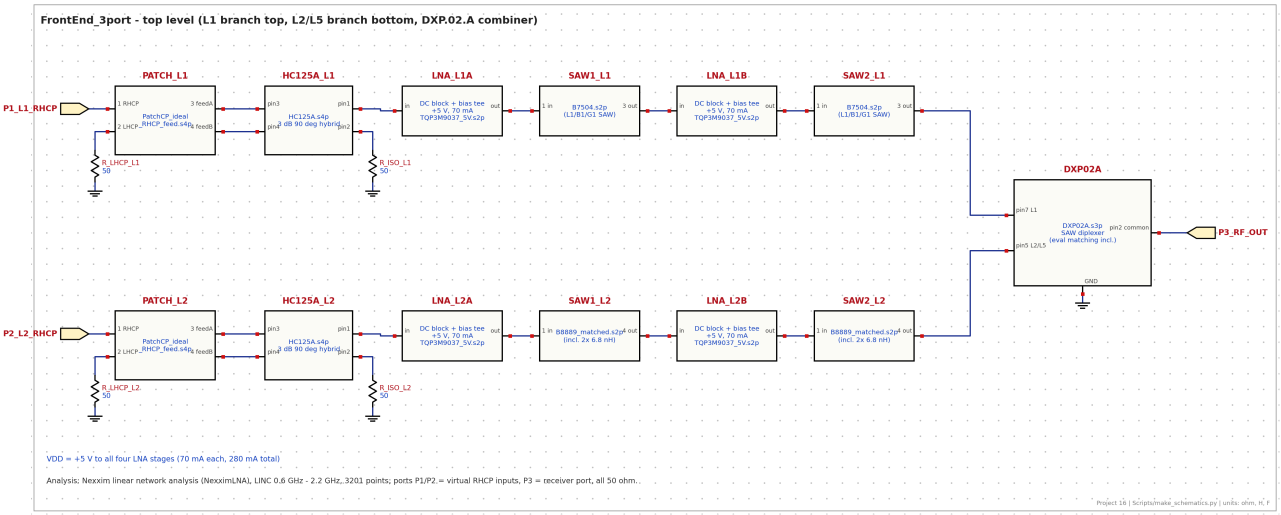


Figure 2 - Top-level schematic, design FrontEnd_3port (as built by Scripts/aedt_circuit.py).

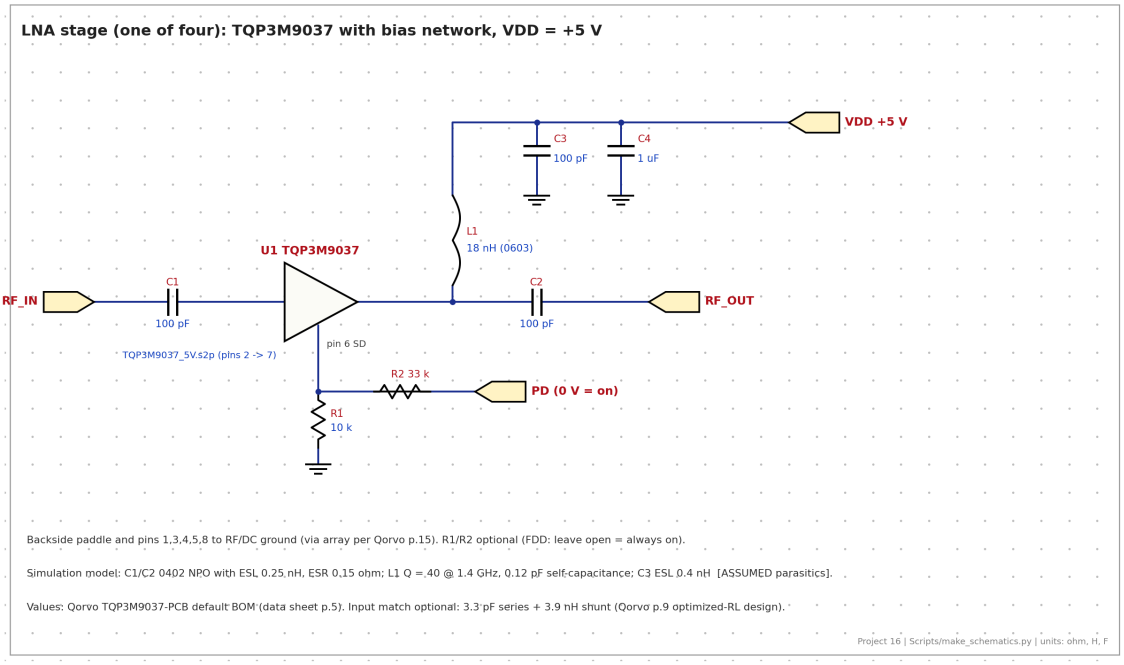


Figure 3 - LNA stage with bias network (one of four).

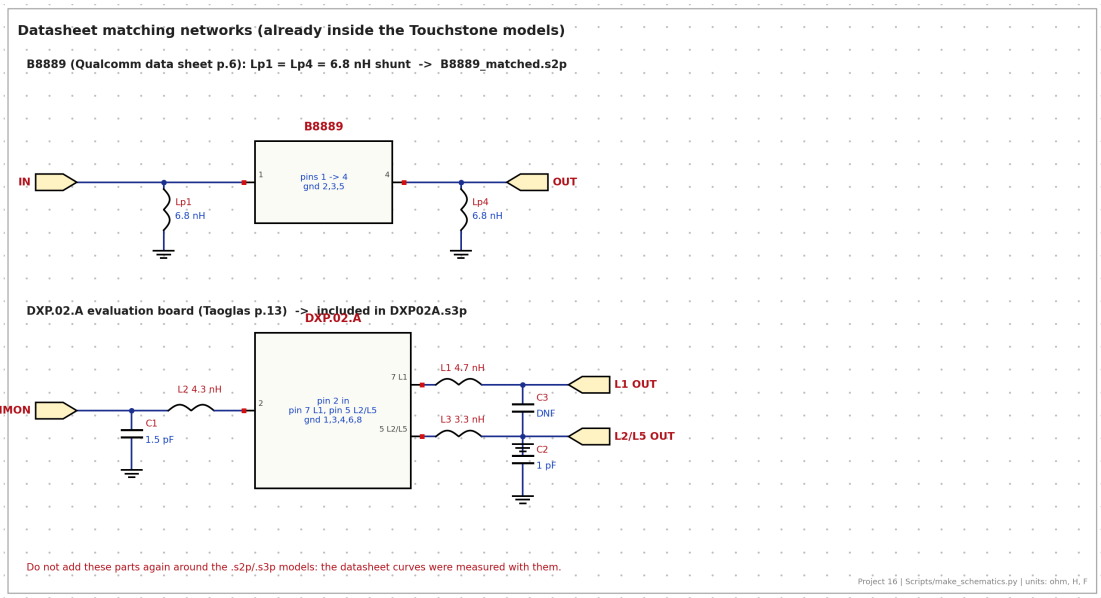


Figure 4 - Datasheet matching networks contained in the SAW and diplexer models.

7 S-parameter results

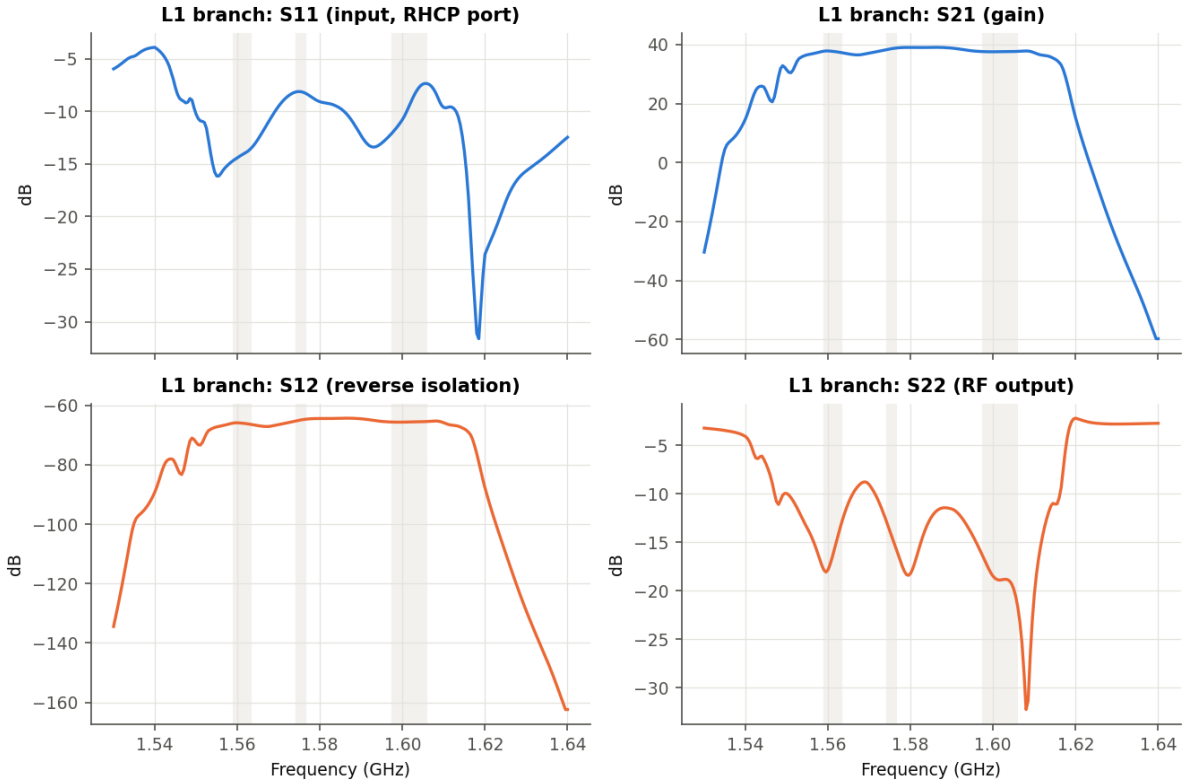


Figure 5 - L1 branch S11 / S21 / S12 / S22 (shaded: B1I, L1/E1, G1).

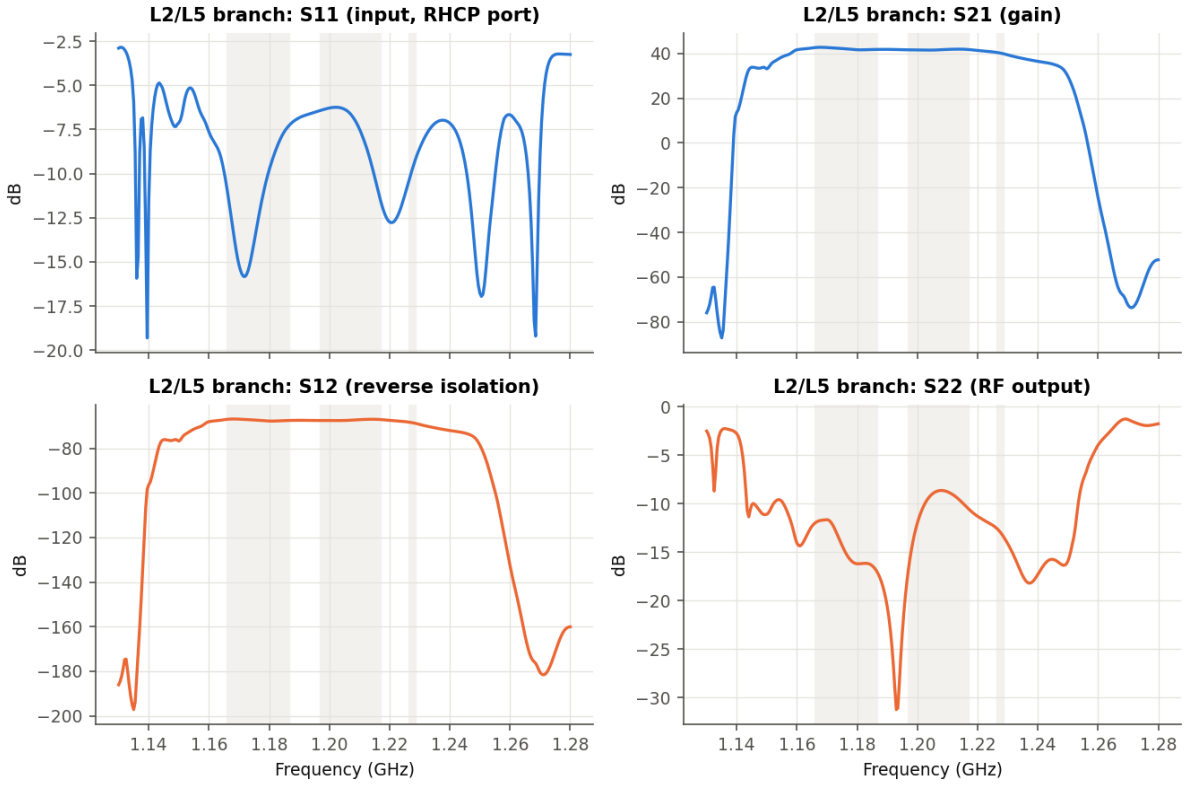


Figure 6 - L2/L5 branch S11 / S21 / S12 / S22 (shaded: L5/E5a, E5b, L2).

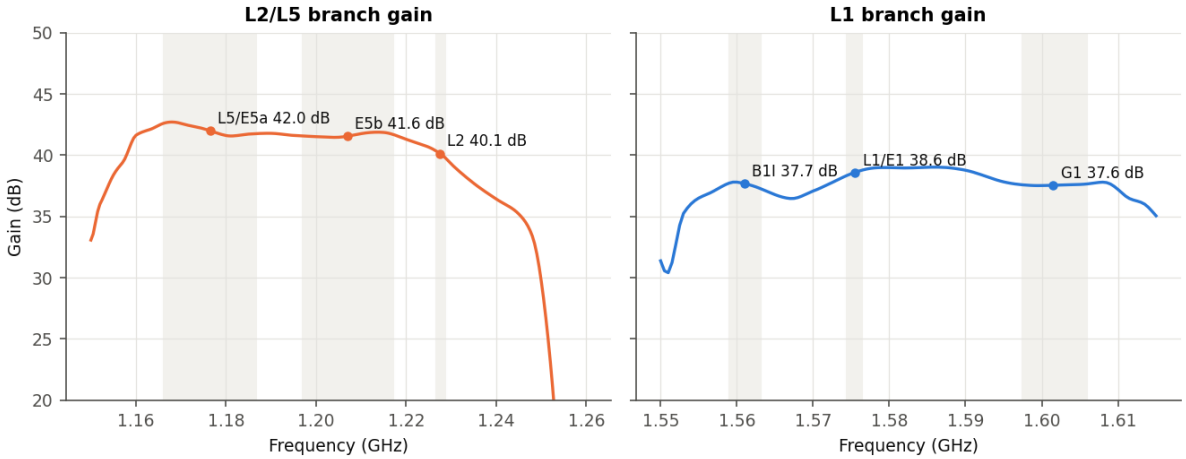


Figure 7 - In-band gain of both branches.

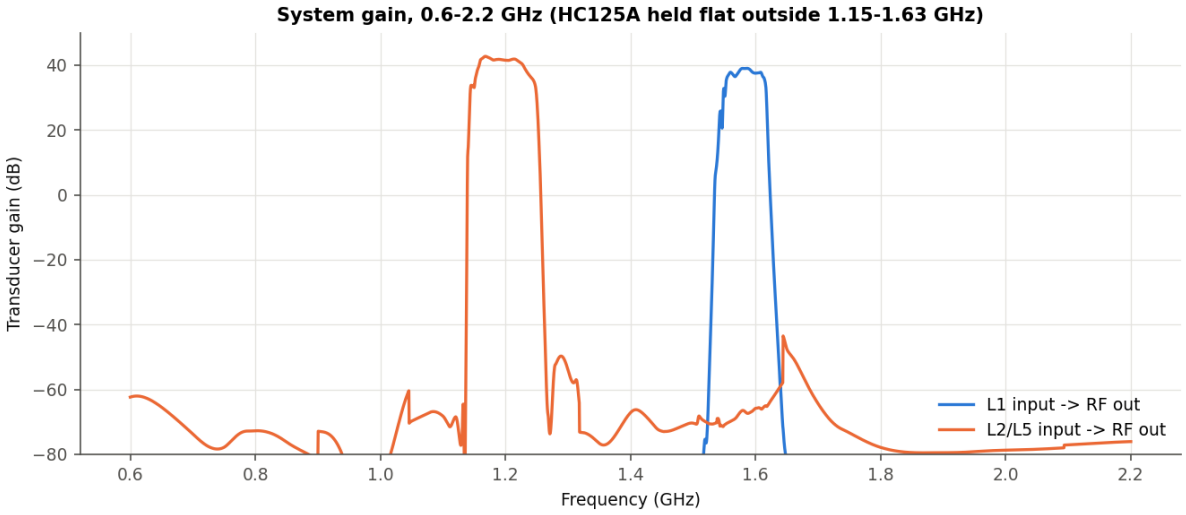


Figure 8 - Wide-band transfer 0.6-2.2 GHz.

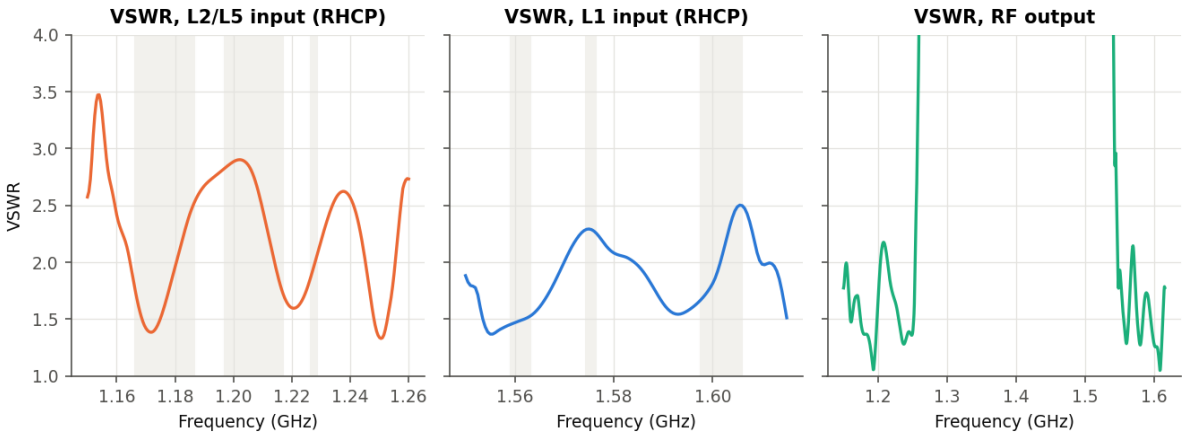


Figure 9 - VSWR at the RHCP inputs and the RF output.

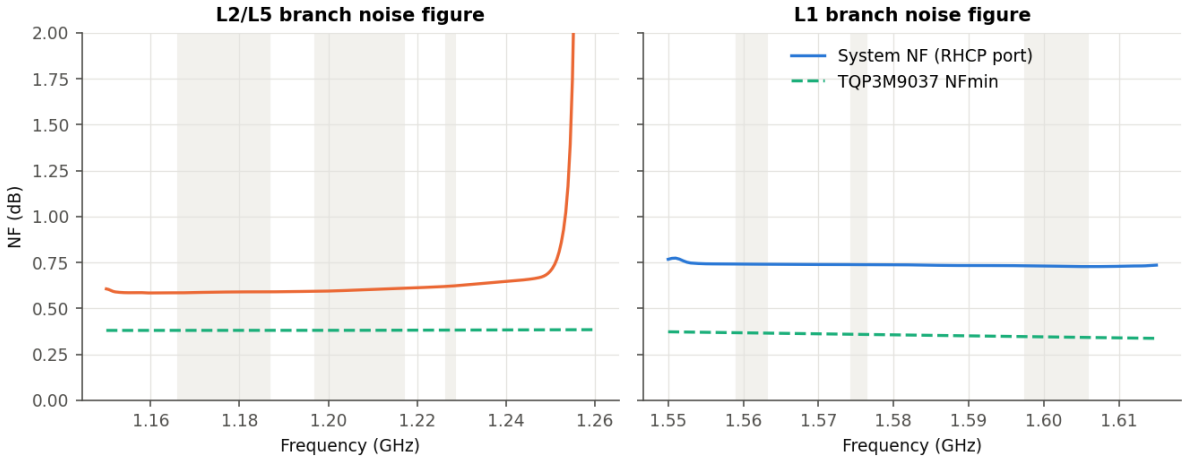


Figure 10 - System noise figure (both branches' output noise included) and LNA NFmin.

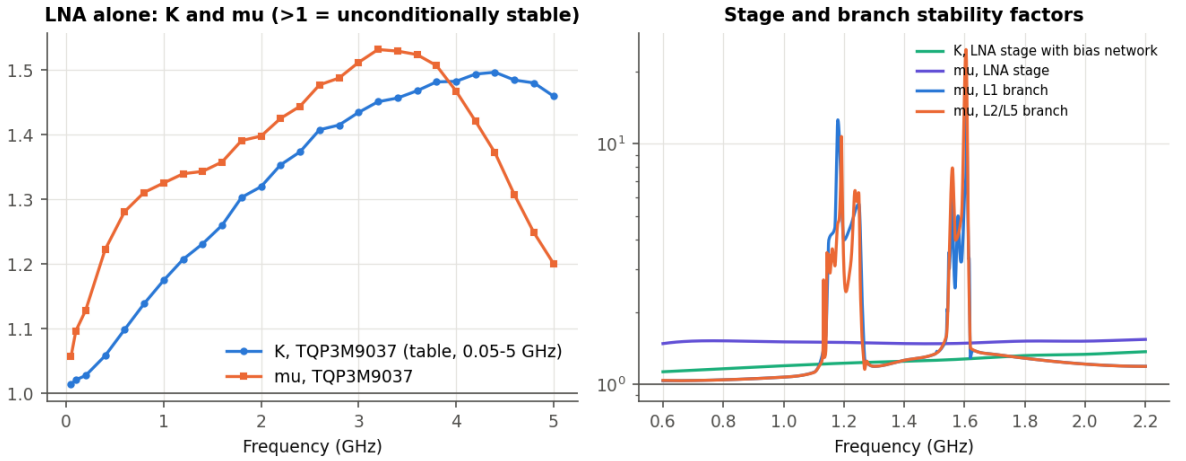


Figure 11 - Stability: K and mu. LNA alone: K_min = 1.014 at 0.05 GHz, mu_min = 1.057, |D|max = 0.71; LNA stage with bias: K_min = 1.13, mu_min = 1.48.

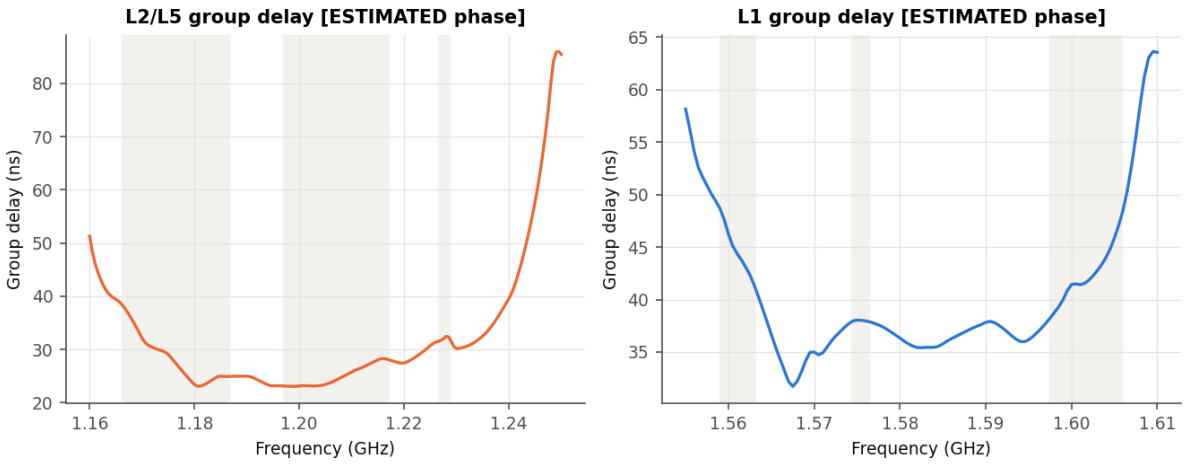


Figure 12 - Group delay (estimated: SAW/DXP phase is minimum-phase).

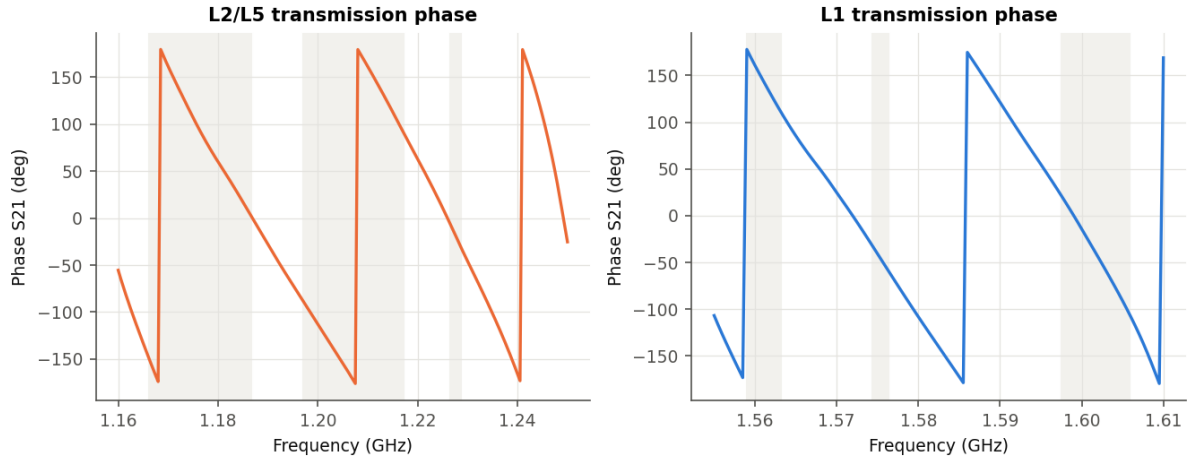


Figure 13 - Transmission phase.

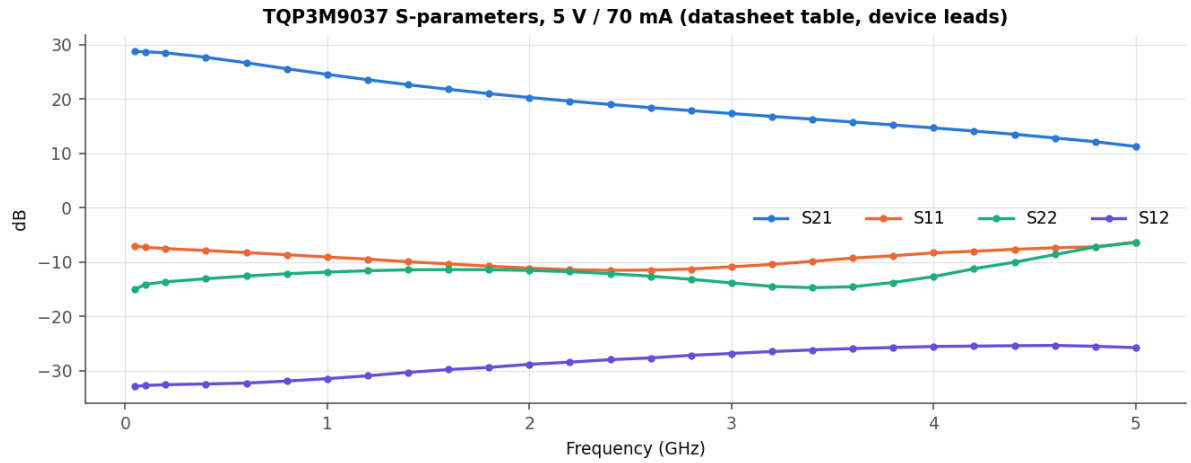


Figure 14 - TQP3M9037 datasheet S-parameters (5 V, 70 mA).

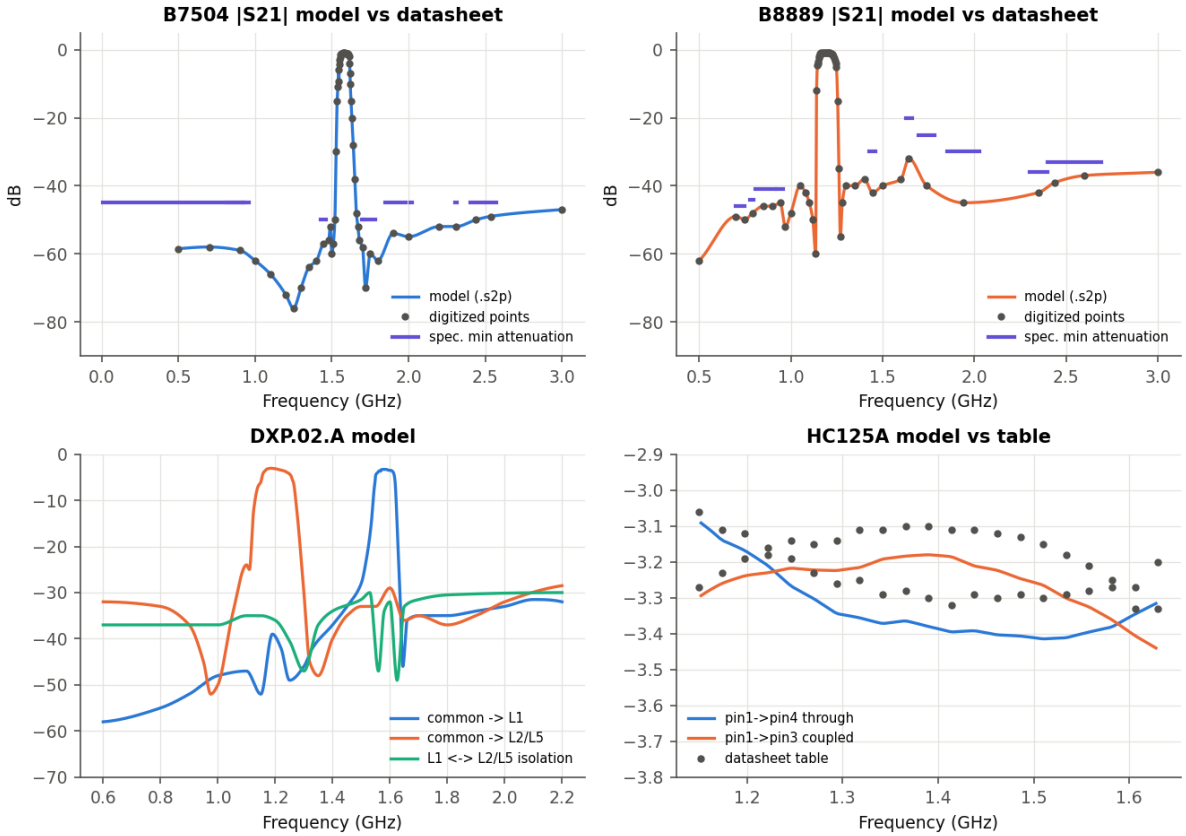


Figure 15 - Component models against the datasheet data.

7A Rejection, insertion loss and other RF parameters

Rejection is quoted in dBc: how far below the band-centre gain a frequency is passed (rejection = G(fc) - G(f)). Filter rejection is shown per component against the datasheet minimum-attenuation table, and for the complete chain. Model rejection beyond ~60-80 dBc is not achievable on a real board: radiation and ground currents leak around the filters, so treat deeper values as 'limited by layout'.

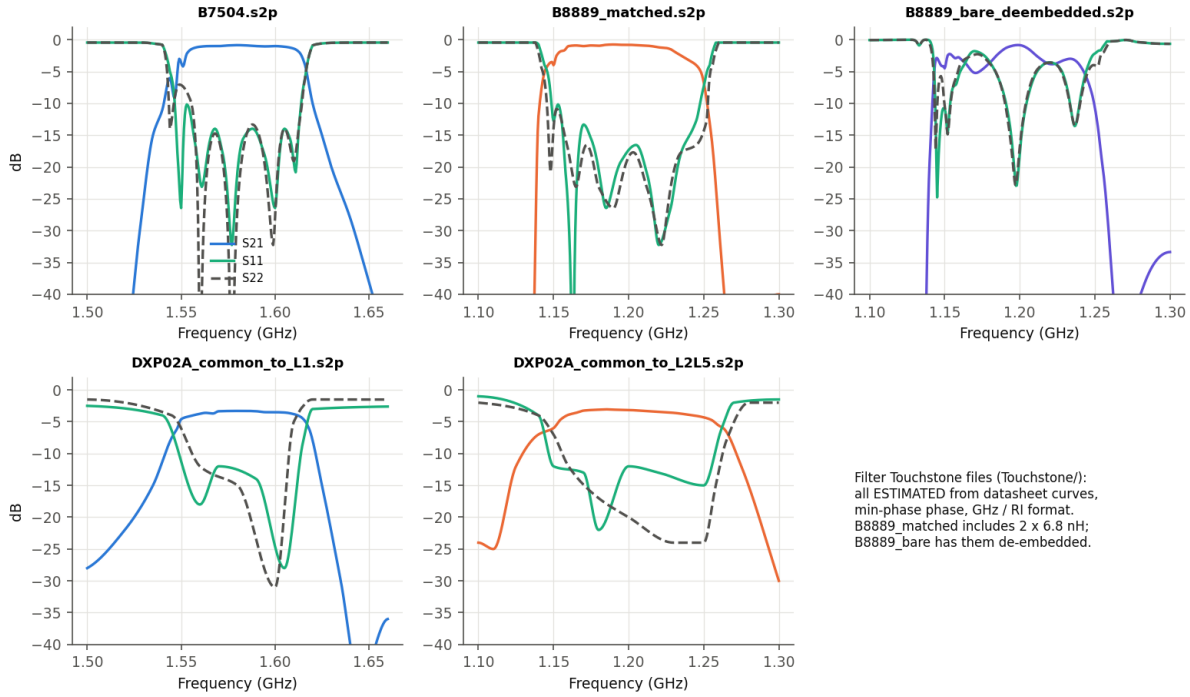


Figure 15-0 - The filter Touchstone files (pass-band view).

File	Points	Range (MHz)	Passive	Use
B7504.s2p	11981	10-6000	yes	L1 SAW, no matching needed (50 ohm)
B8889_matched.s2p	11981	10-6000	yes	L2/L5 SAW with its 2 x 6.8 nH (use as one block)
B8889_bare_deembedded.s2p	11401	300-6000	yes	L2/L5 SAW chip only - add your own matching
DXP02A_common_to_L1.s2p	3201	600-2200	yes	diplexer L1 path, L2/L5 port in 50 ohm
DXP02A_common_to_L2L5.s2p	3201	600-2200	yes	diplexer L2/L5 path, L1 port in 50 ohm
DXP02A.s3p	3201	600-2200	yes	full 3-port diplexer

Table 7A-0 - Filter S-parameter files (all ESTIMATED from datasheet curves).

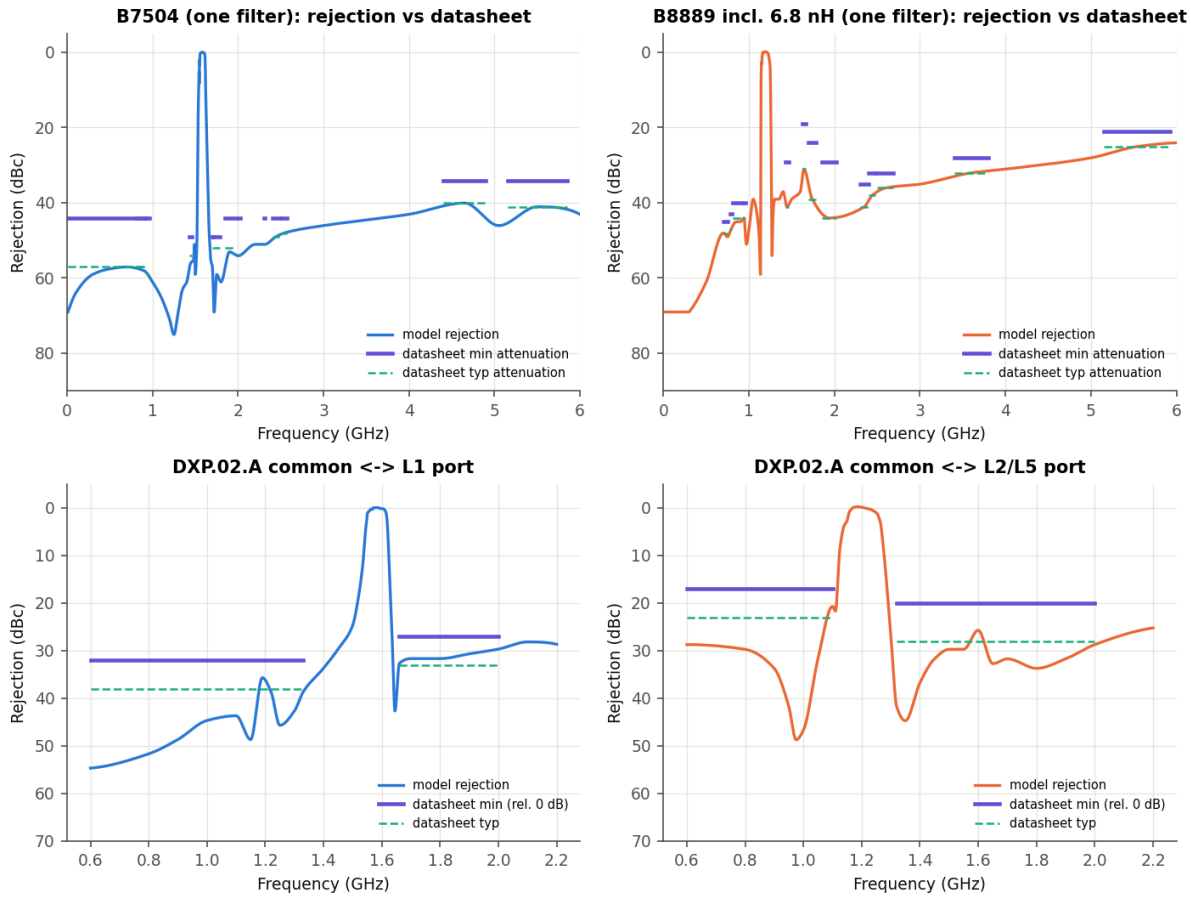


Figure 15A - Rejection of each filter (one device) against the datasheet limits: B7504, B8889 (with its 6.8 nH matching), DXP.02.A common-L1 and common-L2/L5 paths.

Filter	Band (MHz)	Spec min att. (dB)	Spec typ (dB)	Model min att. (dB)	Margin to spec (dB)
B7504	10-925	45	58	58.0	13.0
B7504	814-915	45	58	58.3	13.3
B7504	925-960	45	60	59.4	14.4
B7504	1427-1463	50	55	56.6	6.6
B7504	1695-1710	50	53	57.4	7.4
B7504	1710-1785	50	53	60.0	10.0
B7504	1850-1980	45	53	54.0	9.0
B7504	2010-2025	45	55	54.9	9.9
B7504	2305-2315	45	52	52.0	7.0
B7504	2401-2483	45	50	49.5	4.5
B7504	2500-2570	45	49	48.8	3.8
B7504	4400-4900	35	41	41.0	6.0
B7504	5150-5850	35	42	42.0	7.0
B8889	698-748	46	49	49.0	3.0
B8889	777-798	44	48	47.7	3.7
B8889	807-915	41	45	45.7	4.7
B8889	925-960	41	45	45.0	4.0
B8889	1427-1463	30	42	40.6	10.6
B8889	1626.5-1660.5	20	32	32.0	12.0
B8889	1695-1785	25	40	35.9	10.9
B8889	1850-2025	30	45	43.8	13.8
B8889	2300-2400	36	42	40.3	4.3
B8889	2400-2483	33	39	38.2	5.2
B8889	2500-2690	33	37	36.7	3.7
B8889	3400-3800	29	33	32.5	3.5
B8889	5150-5925	22	26	25.1	3.1
DXP L1	600-1330	32	38	39.0	3.7
DXP L1	1660-2000	27	33	33.0	2.7

Filter	Band (MHz)	Spec min att. (dB)	Spec typ (dB)	Model min att. (dB)	Margin to spec (dB)
DXP L2/L5	600-1100	17	23	24.0	3.7
DXP L2/L5	1320-2000	20	28	29.0	5.7

Table 7A-1 - Filter rejection vs the datasheet attenuation specification (worst point in each band). All bands pass; smallest margins ~3 dB (B8889 cellular bands, DXP).

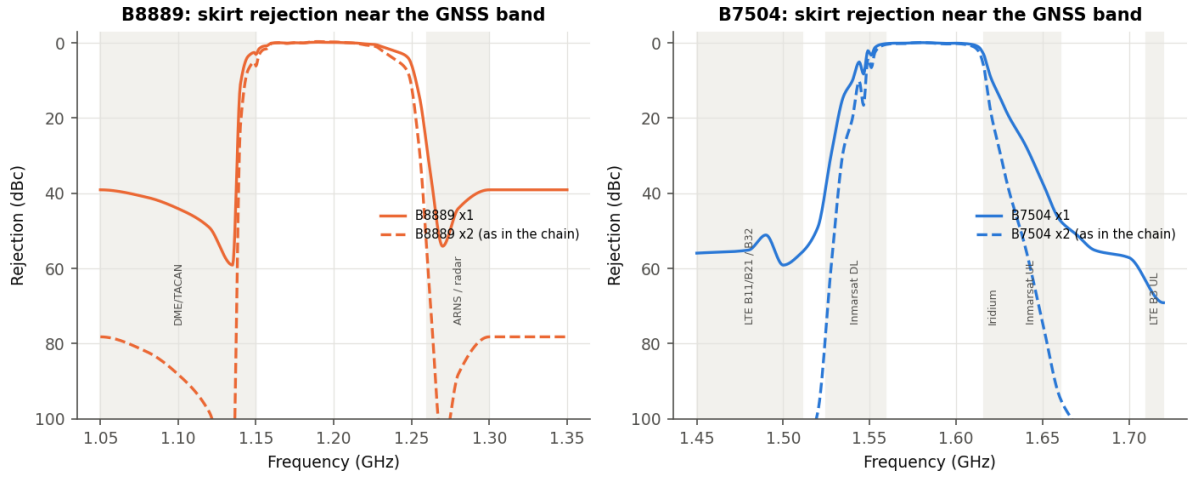


Figure 15B - Skirt rejection next to the GNSS bands, one filter and the two filters in series as used in each branch. The weak points are the B8889 lower skirt (DME, < 4 dB at 1150 MHz per filter) and the B7504 upper skirt (Iridium, ~5-11 dB per filter).

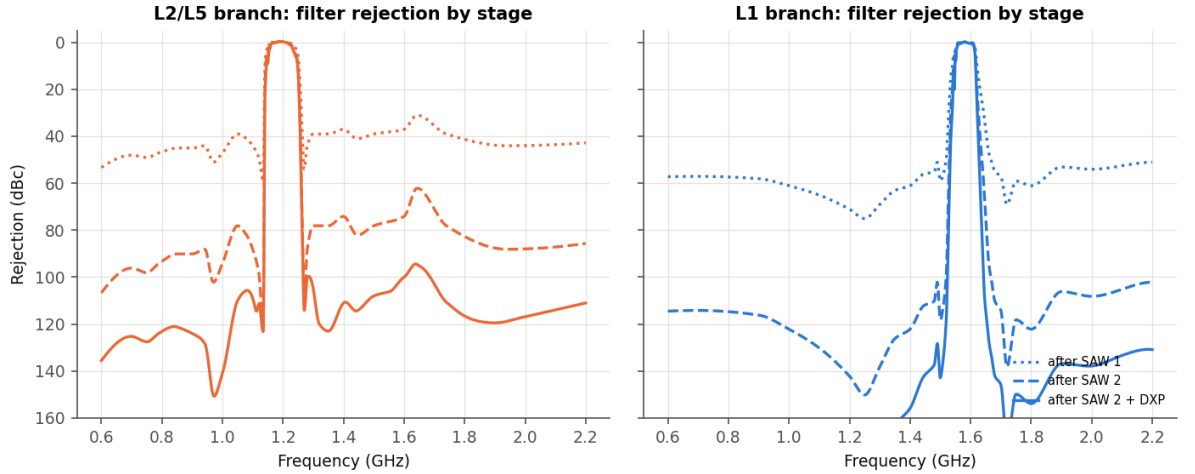


Figure 15C - Cumulative filter rejection: after SAW 1, after SAW 2, after SAW 2 + DXP.

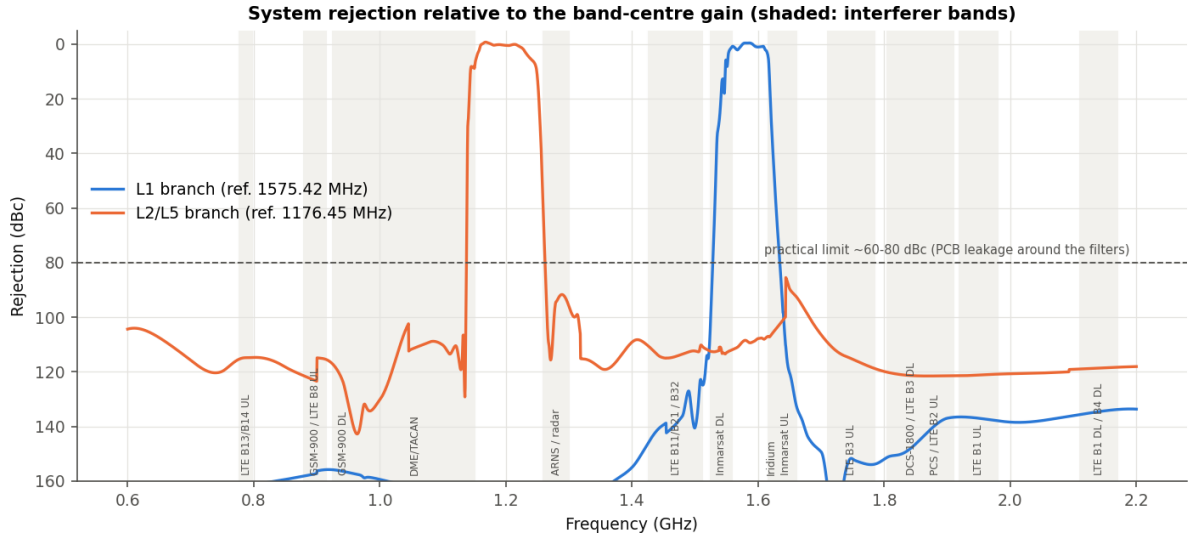


Figure 15D - Rejection of the complete system (LNAs included) with the interferer bands.

Interferer band	MHz	L1 branch min (dBc)	L1 mean (dBc)	L2/L5 branch min (dBc)	L2/L5 mean (dBc)
LTE B13/B14 UL	777-798	161.5	161.9	114.8	114.9
GSM-900 / LTE B8 UL	880-915	155.8	157.0	114.9	119.4
GSM-900 DL	925-960	155.9	156.5	117.1	126.6
DME/TACAN (below L5)	960-1150	157.3	163.2	8.2	111.6
ARNS / radar	1260-1300	168.7	185.2	66.5	96.4
LTE B11/B21 / B32 (L-band)	1427-1511	122.8	137.5	110.2	113.4
Inmarsat DL	1525-1559	0.9	29.6	110.9	112.5
Iridium	1616-1626.5	4.7	28.4	104.8	106.3
Inmarsat UL	1626.5-1660.5	51.6	101.9	85.5	96.2
LTE B3 UL	1710-1785	151.7	159.4	109.2	114.9
DCS-1800 / LTE B3 DL	1805-1880	139.8	147.0	120.2	121.2
PCS / LTE B2 UL	1850-1910	136.7	140.4	121.5	121.5
LTE B1 UL	1920-1980	136.5	137.1	120.9	121.2
LTE B1 DL / B4 DL	2110-2170	133.8	134.7	118.3	118.6

Table 7A-2 - System rejection in the interferer bands (min = worst point in the band). Values above ~80 dBc are model values; expect layout-limited results in practice. Inmarsat DL (1525-1559 MHz) touches the L1 pass-band edge, hence the low minimum.

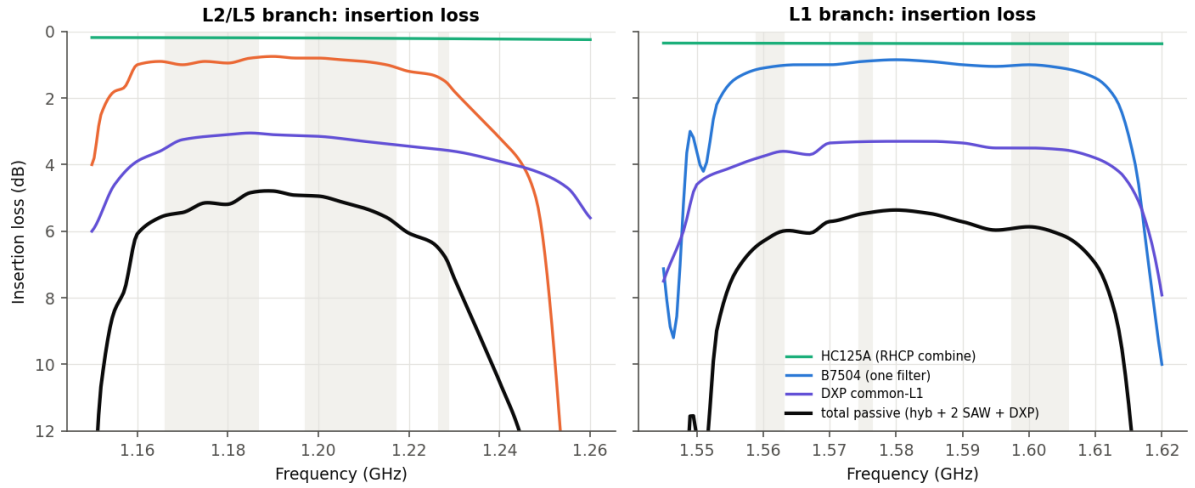


Figure 15E - Insertion loss of each passive part and of the passive chain per branch.

Band	f (MHz)	Hybrid (dB)	SAW, one (dB)	SAW max in BW (dB)	DXP (dB)	Total passive (dB)	Total max in BW (dB)
L1/E1/B1C	1575.42	0.36	0.89	0.91	3.31	5.46	5.49
B1I	1561.10	0.35	1.07	1.12	3.69	6.18	6.37
G1	1602.00	0.37	1.02	1.13	3.51	5.92	6.21
L5/E5a/B2a	1176.45	0.19	0.91	1.00	3.14	5.15	5.52
E5b/B2b	1207.14	0.20	0.87	1.08	3.25	5.19	5.77
L2	1227.60	0.22	1.45	1.56	3.56	6.68	6.91
G2	1248.00	0.24	5.00	9.56	4.20	14.44	23.81

Table 7A-3 - Insertion loss (total passive = hybrid + 2 SAW + DXP; the LNAs make up for it).

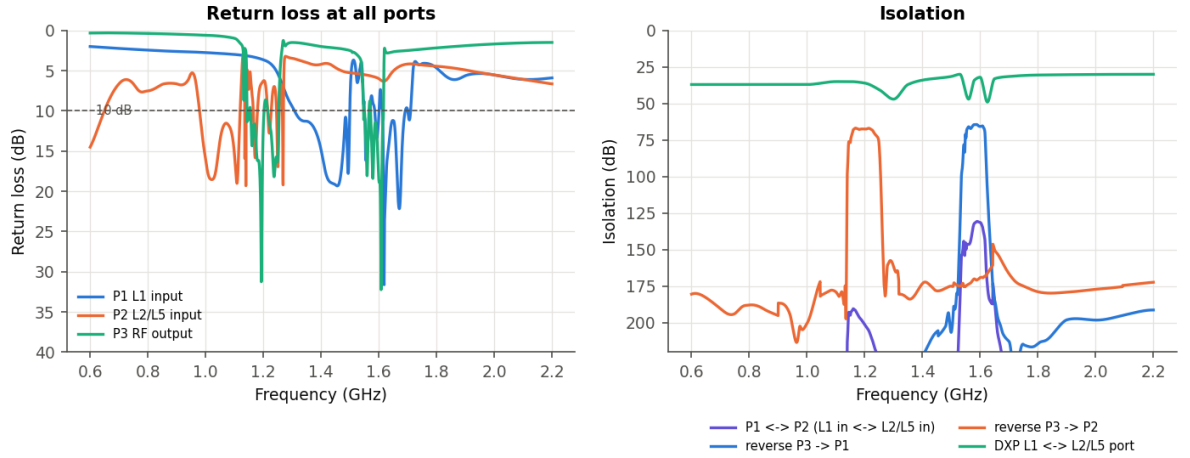


Figure 15F - Return loss at all ports (10 dB line) and isolation: between the two antenna inputs, reverse isolation output-to-input, and the DXP port-to-port isolation.

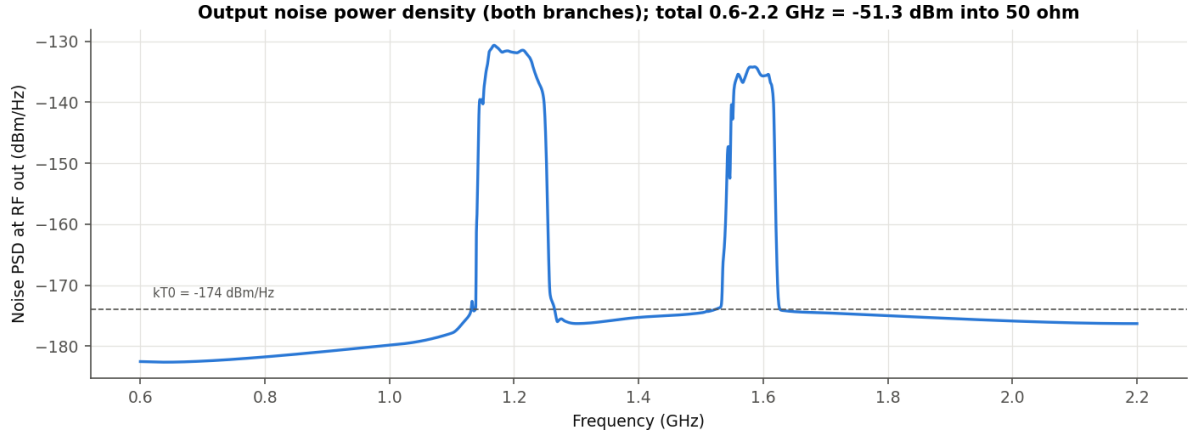


Figure 15G - Output noise density delivered to the receiver. Total noise power 0.6-2.2 GHz = -51.3 dBm (the receiver front end must handle it without compressing).

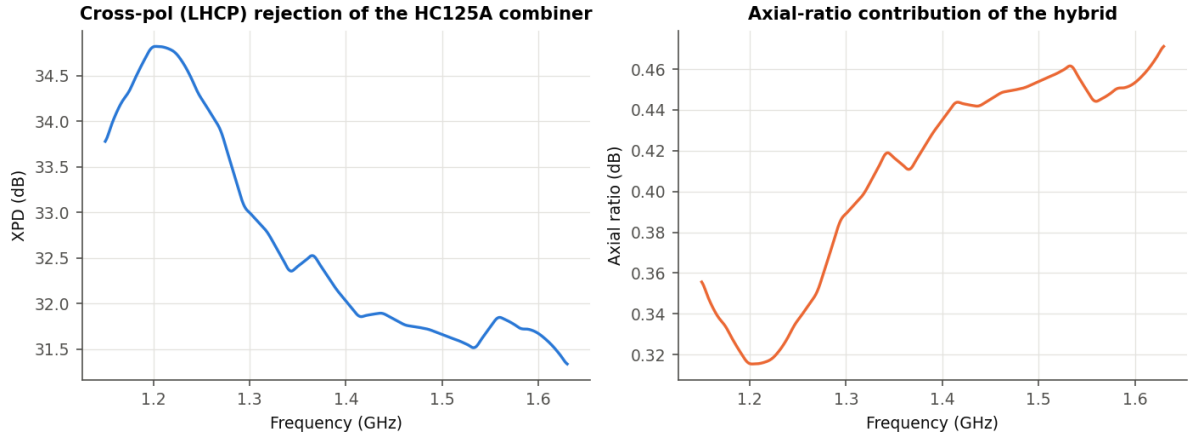


Figure 15H - Cross-polar (LHCP) rejection and axial-ratio contribution caused by the HC125A amplitude/phase imbalance (ideal patch assumed; the real antenna axial ratio adds to this).

Band	Gain	NF	Te (K)	RL in	RL out	Mismatch loss in	Rev. iso	XPD	AR (dB)	GD (ns)	IIP3 / OIP3 (dBm)	IP1dB in / OP1dB out (dBm)	SFDR (dB)	C/N0 (dB-Hz)
L1/E1/B1C	38.6	0.74	54	8.1	14.4	0.72	65	31.8	0.45	38.0	-7.9 / 30.7	-20.8 / 16.8	68.2	44.8
B1I	37.7	0.74	54	14.2	16.6	0.17	66	31.8	0.44	44.4	-	-	-	-
G1	37.6	0.73	53	9.2	18.9	0.56	66	31.7	0.45	41.8	-	-	-	-

Band	Gain	NF	Te (K)	RL in	RL out	Mismatch loss in	Rev. iso	XPD	AR (dB)	GD (ns)	IIP3 / OIP3 (dBm)	IP1dB in / OP1dB out (dBm)	SFDR (dB)	C/N0 (dB-Hz)
L5/E5a/B2a	42.0	0.59	42	12.5	15.2	0.25	67	34.4	0.33	27.3	-11.6 / 30.4	-23.6 / 17.4	59.2	45.5
E5b/B2b	41.6	0.60	43	6.6	8.7	1.07	67	34.8	0.32	24.2	-	-	-	-
L2	40.1	0.62	45	9.8	12.9	0.48	68	34.7	0.32	32.0	-10.5 / 29.6	-22.7 / 16.5	66.5	43.4
G2	33.4	0.67	48	13.7	16.3	0.19	75	34.3	0.34	80.1	-	-	-	-

Table 7A-4 - RF parameter summary per GNSS band (dB unless stated). SFDR in the signal bandwidth (2.046 MHz L1/L2, 20.46 MHz L5); C/N0 for the minimum signal with the antenna at 290 K.

Inter-frequency group delay: L1 - L5 = 10.7 ns through the front end (estimated, min-phase SAW models). Dual-frequency receivers calibrate this as an inter-frequency bias; it must be stable over temperature, which SAW filters are not perfectly (not modelled).

Band	Gain typ (dB)	Gain min (dB)	Gain max (dB)	NF typ (dB)	NF max (dB)
L1	37.9	33.2	41.8	0.77	0.97
L5	41.0	35.7	45.3	0.60	0.97
L2	39.6	33.1	44.0	0.63	0.98

Table 7A-5 - Worst case from the datasheet limits (Friis, 50 ohm): LNA gain typ +/-1.5 dB and NF max 0.65 dB, SAW and DXP IL max, hybrid IL max 0.3 dB. The receiver must accept the full gain range.

8 Cascade analysis and power budget

Stage values at band centre in 50 ohm (passives: NF = loss at T0). Signal = minimum received power for a 0 dBic antenna. Noise = kT0 B F G in the signal bandwidth; 'C/N' is the ratio in that bandwidth (negative for GNSS - the signal is recovered by despreading; C/N0 = C/N + 10 log10 B).

Stage	G (dB)	NF (dB)	Cum G	Cum NF	Cum IIP3 (dBm)	Cum IP1dB (dBm)	Signal (dBm)	Noise (dBm)	C/N (dB)
HC125A (RHCP combine)	-0.36	0.36	-0.36	0.36	-	-	-128.9	-110.9	-18.0
LNA1 TQP3M9037 + bias	21.95	0.37	21.59	0.73	13.2	0.3	-106.9	-88.5	-18.4
B7504 #1	-0.89	0.89	20.70	0.74	13.2	0.3	-107.8	-89.4	-18.4
LNA2 TQP3M9037 + bias	21.95	0.37	42.65	0.74	-7.9	-20.8	-85.8	-67.5	-18.4
B7504 #2	-0.89	0.89	41.76	0.74	-7.9	-20.8	-86.7	-68.4	-18.4
DXP.02.A (combiner)	-3.31	3.31	38.45	0.74	-7.9	-20.8	-90.0	-71.7	-18.4

Cascade - L1 C/A (1575.42 MHz, 2.046 MHz).

Stage	G (dB)	NF (dB)	Cum G	Cum NF	Cum IIP3 (dBm)	Cum IP1dB (dBm)	Signal (dBm)	Noise (dBm)	C/N (dB)
HC125A (RHCP combine)	-0.19	0.19	-0.19	0.19	-	-	-128.1	-100.9	-27.2
LNA1 TQP3M9037 + bias	23.61	0.39	23.41	0.58	11.2	-0.9	-104.5	-76.9	-27.6
B8889 #1	-0.91	0.91	22.50	0.59	11.2	-0.9	-105.4	-77.8	-27.6
LNA2 TQP3M9037 + bias	23.61	0.39	46.11	0.59	-11.6	-23.6	-81.8	-54.2	-27.6
B8889 #2	-0.91	0.91	45.20	0.59	-11.6	-23.6	-82.7	-55.1	-27.6
DXP.02.A (combiner)	-3.14	3.14	42.06	0.59	-11.6	-23.6	-85.8	-58.2	-27.6

Cascade - L5 (1176.45 MHz, 20.46 MHz).

Stage	G (dB)	NF (dB)	Cum G	Cum NF	Cum IIP3 (dBm)	Cum IP1dB (dBm)	Signal (dBm)	Noise (dBm)	C/N (dB)
HC125A (RHCP combine)	-0.22	0.22	-0.22	0.22	-	-	-130.2	-110.9	-19.4
LNA1 TQP3M9037 + bias	23.39	0.40	23.17	0.61	11.4	-0.7	-106.8	-87.1	-19.7
B8889 #1	-1.45	1.45	21.71	0.62	11.4	-0.7	-108.3	-88.5	-19.8
LNA2 TQP3M9037 + bias	23.39	0.40	45.10	0.62	-10.5	-22.7	-84.9	-65.1	-19.8
B8889 #2	-1.45	1.45	43.65	0.62	-10.5	-22.7	-86.4	-66.6	-19.8
DXP.02.A (combiner)	-3.56	3.56	40.09	0.62	-10.5	-22.7	-89.9	-70.2	-19.8

Cascade - L2C (1227.60 MHz, 2.046 MHz).

L1 C/A: C/N0 at the RF output = 44.7 dB-Hz for the minimum signal (sky noise at 290 K assumed; with a real sky temperature of ~100 K the front-end NF matters more and C/N0 rises by ~2-3 dB). The Friis cascade (50 ohm stages) and the full noise-correlation simulation agree within 0.02 dB.

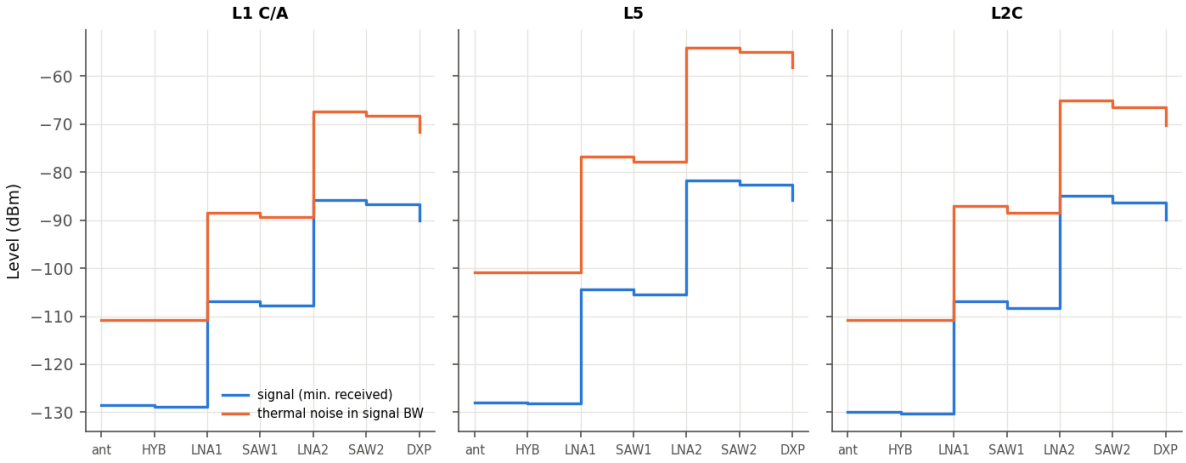


Figure 16 - Level diagram: minimum signal and thermal noise through the chain.

Power consumption: 4 x 70 mA x 5 V = 1.40 W (VDD 4.2 V typ allowed: 4 x ~60 mA). DC is fed through the 18 nH chokes; if the antenna is powered over the coax, add a bias tee at the RF output (the DXP and SAWs block DC; the SAWs are rated 5 V DC max).

9 Interference and linearity

Out-of-band signals at the antenna feed (levels assumed for a co-located terminal, A18) propagated through each branch with the small-signal gains. 'Back-off' = LNA OP1dB minus the interferer level at that LNA output.

Interferer	f (MHz)	P ant (dBm)	Branch	LNA1 out	SAW1 out	LNA2 out	RF out	Back-off LNA1	Back-off LNA2
LTE B13/B14 UL	787.0	-10	L1	15.0	-43.2	-18.0	-131.5	7.0	40.0
LTE B13/B14 UL	787.0	-10	L2/L5	15.0	-33.1	-7.9	-88.9	7.0	29.9
GSM-900 / LTE B8 UL	897.5	-10	L1	14.6	-44.4	-19.6	-130.7	7.4	41.6
GSM-900 / LTE B8 UL	897.5	-10	L2/L5	14.6	-31.4	-6.6	-89.5	7.4	28.6
DME/TACAN (aviation)	1150.0	-30	L1	-6.5	-75.2	-51.5	-172.2	28.1	73.1
DME/TACAN (aviation)	1150.0	-30	L2/L5	-6.5	-10.5	13.3	3.3	28.1	8.3
ARNS radar	1260.0	-40	L1	-17.0	-92.6	-69.3	-193.7	38.4	90.7
ARNS radar	1260.0	-40	L2/L5	-17.0	-46.1	-22.8	-57.5	38.4	44.3
Inmarsat L-band DL	1545.0	-80	L1	-58.3	-65.4	-43.3	-58.0	79.2	64.3
Inmarsat L-band DL	1545.0	-80	L2/L5	-58.3	-97.5	-75.4	-147.6	79.2	96.4
Iridium	1621.0	-20	L1	1.4	-9.7	12.0	-8.3	19.4	8.8
Iridium	1621.0	-20	L2/L5	1.4	-33.5	-11.7	-78.3	19.4	32.5
Inmarsat UL	1643.0	-20	L1	1.3	-29.5	-7.8	-83.5	19.5	28.6
Inmarsat UL	1643.0	-20	L2/L5	1.3	-30.7	-9.0	-76.6	19.5	29.8
LTE B3 UL	1747.5	-10	L1	10.9	-49.3	-28.0	-123.2	9.7	48.6
LTE B3 UL	1747.5	-10	L2/L5	10.9	-29.4	-8.1	-84.3	9.7	28.7
DCS-1800 DL	1842.5	-30	L1	-9.4	-68.3	-47.4	-141.0	29.7	67.7
DCS-1800 DL	1842.5	-30	L2/L5	-9.4	-53.0	-32.1	-112.2	29.7	52.4
LTE B1 UL	1950.0	-10	L1	10.2	-44.3	-23.8	-111.8	9.8	43.8
LTE B1 UL	1950.0	-10	L2/L5	10.2	-34.8	-14.3	-92.7	9.8	34.3

Table - Interferer levels in dBm at each node.

- DME/TACAN 960-1215 MHz (pulsed, aviation): the B8889 lower skirt gives only ~4 dB at 1150 MHz, so LNA 2 output reaches +13.3 dBm and the RF output +3.3 dBm for -30 dBm at the antenna. This exceeds the B8889 +5 dBm input rating at SAW 2 and can block the receiver. Critical for aviation/automotive near airports.
- Iridium 1616-1626.5 MHz: B7504 attenuation at 1621 MHz ~11 dB per filter; LNA 2 output +12.0 dBm for -20 dBm at the antenna.
- Cellular uplinks (-10 dBm): LNA 1 sees the full level (no pre-filter); back-off 7-10 dB from OP1dB, ~13 dB below the +22 dBm absolute-maximum input. After SAW 1 the levels drop by 40-60 dB.
- In-band: cascaded input P1dB -21 to -24 dBm and IIP3 -8 to -12 dBm, limited by LNA 2 (the GNSS signal itself is ~100 dB below this).

10 Simulation tool cross-checks

Solver	How	Max |dS| in band vs reference	Script
scikit-rf 1.x (reference)	Network cascade + Circuit, noise-correlation matrices	-	Scripts/simulate.py
heavi 0.7.9 (nodal Y solver)	Same circuit from the Touchstone files, own R/L/C	0.0037 dB	Scripts/crosscheck_heavi.py
circuitRF 1.6.1 (CLI sparam)	Netlist circuitRF/gnss_frontend.cnl, SnP blocks	0.0007 dB	Scripts/build_circuitrf.py
Qucsator-RF 1.0.7 (Qucs-S 26.1.1)	Netlist QucsS/gnss_frontend.net, SPfile polar/cubic	0.0007 dB	Scripts/build_qucs.py

Table - Solver agreement on S31, S32, S11, S22, S33 inside 1164-1230 and 1559-1606 MHz.

ANSYS AEDT: the PyAEDT script builds, solves and saves the project. In this session AEDT 2026 R1 is installed but the headless gRPC session could not be started from the automation shell (see the error log, appendix A); run 'python Scripts/aedt_circuit.py' on the workstation (add --gui to watch it). The script exports ANSYS_Project/AEDT_System_3port.s3p, which Scripts/build_all.py then compares with the reference.

Sensitivity found during the cross-check: interpolating the 200 MHz-step LNA table linearly instead of with a cubic spline changes in-band gain by up to 1.4 dB. The LNA table resolution is therefore a model uncertainty of the order of +/-0.7 dB in gain (assumption A6).

11 Model validation against the datasheets

Item	Datasheet typ	Datasheet max	Model
B7504 IL 1559.1-1563.1 MHz	1.10	2.00	1.06
B7504 IL 1574.4-1576.4 MHz	0.90	1.40	0.90
B7504 IL 1597.5-1605.9 MHz	1.20	2.00	1.04
B8889 IL 1166.22-1186.68 MHz	0.90	1.50	0.92
B8889 IL 1205.09-1209.19 MHz	1.00	1.30	0.87
B8889 IL 1226.58-1228.62 MHz	1.40	2.60	1.48
B7504 att 10-925 MHz (min 45)	58.00	-	58.00
B7504 att 814-915 MHz (min 45)	58.00	-	58.34
B7504 att 925-960 MHz (min 45)	60.00	-	59.44
B7504 att 1427-1463 MHz (min 50)	55.00	-	56.58
B7504 att 1695-1710 MHz (min 50)	53.00	-	57.37
B7504 att 1710-1785 MHz (min 50)	53.00	-	60.00
B7504 att 1850-1980 MHz (min 45)	53.00	-	54.00
B7504 att 2010-2025 MHz (min 45)	55.00	-	54.87
B7504 att 2305-2315 MHz (min 45)	52.00	-	51.99
B7504 att 2401-2483 MHz (min 45)	50.00	-	49.48
B7504 att 2500-2570 MHz (min 45)	49.00	-	48.77
B7504 att 4400-4900 MHz (min 35)	41.00	-	41.00
B7504 att 5150-5850 MHz (min 35)	42.00	-	42.00
B8889 att 698-748 MHz (min 46)	49.00	-	49.00
B8889 att 777-798 MHz (min 44)	48.00	-	47.66
B8889 att 807-915 MHz (min 41)	45.00	-	45.71
B8889 att 925-960 MHz (min 41)	45.00	-	45.00
B8889 att 1427-1463 MHz (min 30)	42.00	-	40.59
B8889 att 1626.5-1660.5 MHz (min 20)	32.00	-	32.00
B8889 att 1695-1785 MHz (min 25)	40.00	-	35.90
B8889 att 1850-2025 MHz (min 30)	45.00	-	43.78
B8889 att 2300-2400 MHz (min 36)	42.00	-	40.35
B8889 att 2400-2483 MHz (min 33)	39.00	-	38.25
B8889 att 2500-2690 MHz (min 33)	37.00	-	36.67
B8889 att 3400-3800 MHz (min 29)	33.00	-	32.47
B8889 att 5150-5925 MHz (min 22)	26.00	-	25.07
B7504 group-delay ripple 1597.5-1605.9 MHz [ns]	4.00	15.00	3.52
DXP IL band 1 @1582.19 MHz	3.80	4.60	3.30
DXP IL band 2 @1206.92 MHz	4.20	5.20	3.25
TQP3M9037 gain @2.0 GHz (table) vs spec @1.95	20.00	21.50	20.28

Table - Model vs datasheet (insertion loss: mean over the band; attenuation: worst case in the band).

Every B7504/B8889 attenuation value meets the datasheet minimum and is within ~4 dB of the typical value. The DXP model gives 3.3 dB at the band centres (the measured plots) against 3.8/4.2 dB typical in the table - the table values appear to be band-edge figures (inconsistency I17).

12 Bill of materials

Qty	Ref	Part / value	Package	Manufacturer / note
2	ANT1, ANT2	Dual-feed patch, L1 and L2/L5 (or stacked patch)	-	Not supplied - e.g. Taoglas GPDF series
2	HY1, HY2	HC125A 3 dB hybrid	6.35 x 5.08 x 1.5 mm	Taoglas
2	R1, R2	50 ohm (49.9 ohm 1 %) hybrid isolated-port load	0402	thin film, RF rated
4	U1-U4	TQP3M9037	DFN 2 x 2 mm, 8 pin	Qorvo
8	C1/C2 per LNA	100 pF NPO/C0G 50 V (DC block)	0402	Murata GJM/GRM series
4	C3	100 pF NPO (VDD bypass)	0402	-
4	C4	1.0 uF X5R 10 V (VDD bypass)	0402	-
4	L1	18 nH ceramic, 5 % (RF choke)	0603	e.g. Coilcraft 0603CS / Murata LQW18
8 (opt.)	R1/R2 per LNA	10 k / 33 k shutdown divider	0402	only if PD control is used
2	F1, F2	B7504 SAW (B39162B7504L210)	0.9 x 0.7 mm	Qualcomm RF360
2	F3, F4	B8889 SAW (B39122B8889P810)	1.1 x 0.9 mm	Qualcomm RF360

Qty	Ref	Part / value	Package	Manufacturer / note
4	Lp1/Lp4	6.8 nH wire-wound, high Q (B8889 matching)	0402	e.g. Murata LQW15AN6N8
1	DX1	DXP.02.A SAW diplexer	5 x 5 x 1.7 mm	Taoglas
1	L2	4.3 nH (DXP common)	0402	TDK (eval board)
1	C1	1.5 pF (DXP common)	0402	Murata (eval board)
1	L1	4.7 nH (DXP L1 port)	0402	TDK (eval board)
1	L3	3.3 nH (DXP L2/L5 port)	0402	TDK (eval board)
1	C2	1 pF (DXP L2/L5 port)	0402	Murata (eval board)
1	J1	RF output connector (SMA/U.FL/MMCX)	-	per enclosure

Table - Bill of materials (eval-board reference values).

13 Datasheet inconsistencies and missing parameters

ID	Where	Finding
I1	B7504 vs product brief	Datasheet p.7 lists 'Group delay typ 4.0 / max 15 ns' (GLONASS band); the product brief lists the same numbers as 'group delay ripple'. The model gives ~13 ns absolute delay and ~4-10 ns ripple, so the datasheet label should read 'ripple'.
I2	DXP.02.A front page	Lists 'L2 1227.625 MHz'; GPS L2 is 1227.60 MHz.
I3	DXP.02.A isolation row	'Isolation (1574.22~1576.62 dB)' - unit should be MHz.
I4	DXP.02.A pass-band claim	Text p.11: L2 pin covers '1164-1300 MHz'. Measured plot p.8: insertion loss >6 dB above ~1263 MHz, so GLONASS G2 upper half, B3 (1268.5) and E6 (1278.75) are not usable.
I5	DXP.02.A return loss	Table p.4 band-2 RL typ -8 dB (max -6) vs. the plot p.6 showing 12-24 dB in the L2/L5 band; the plot is used.
I6	DXP.02.A pin naming	Pin 5 is called 'GPS L2 Output' in the pin table but 'GPS L2/L5 OUT' on the symbol.
I7	DXP.02.A RL sign	Return loss given as negative numbers with 'Max -6.0' - ambiguous min/max sense.
I8	HC125A phase balance	Spec says 'Phase balance 90 degrees' with no tolerance; the typical table shows 91.9-93.0 deg (up to 3 deg off), which costs ~0.01 dB of axial-ratio/combining loss.
I9	HC125A pin table vs table columns	Typical table uses 'Coupling/Transmission' and S11..S44 without stating which pins; the pin table uses 'Pin 1..4'. Mapping assumed (A9).
I10	TQP3M9037 noise table	NFmin 0.40 dB at 1.5 GHz but 0.29 dB at 1.7 GHz (non-monotonic); Gopt angle jumps 85.8 -> 42.1 -> 157.4 deg between 2.4/2.5/2.6 GHz with |Gopt| ~0.02-0.08 (measurement scatter).
I11	TQP3M9037 switching speed	Table p.2 note: switching speed with C1 = 10 pF; eval board default C1 = 100 pF (switching table p.5 shows both).
I12	TQP3M9037 stability	'Unconditionally stable' claim holds (K >= 1.01, mu >= 1.06 from the table) but the margin is only ~1 % at 50 MHz.
I13	B8889 spec typo	Data sheet p.7 'Insertion attenuation' first row has no unit; 'insertion loss' typo on B7504 p.7.
I14	B8889 vs brief	Brief gives B8889 pass-band 1166-1228 MHz (62 MHz); the datasheet spec bands end at 1228.62 MHz. GLONASS G2 (1246-1256) is outside the B8889 pass-band (>4 dB at 1245 MHz) although the HC125A table and the DXP claim to cover it.
I15	SDST vs. parts used	The automotive SAW selection table (Feb 2020) does not list B7504 or B8889; equivalent automotive parts are e.g. B39162B2611P810 (L1, 0.9 dB, pre-LNA) and B39122B2620P810 (E5b/L2).
I17	DXP.02.A insertion loss	Table p.4: IL typ 3.8 dB (L1) / 4.2 dB (L2/L5); the measured plots p.7-8 show 3.3 dB / 3.1-3.2 dB at the band centres and reach 3.8 / 4.2 dB only at the band edges. The model follows the plots.
I18	TA1442A bandwidth	Cover page 'BW = 5 MHz' is the specified band 2489.5-2494.5 MHz; the measured 3 dB bandwidth is ~62 MHz (about 2453-2537 MHz) and the filter passes Wi-Fi channel 13 and LTE B41/B7 almost unattenuated.
I19	TA1442A VSWR	Spec VSWR typ 1.2 vs VNA markers 1.04-1.12 in the spec band (sample better than typ).
I20	C2327J5003AHF note	Spec table footnote: 'values are for the case with shunt capacitor across differential lines' - not explained for a 4-port hybrid; no value given.
I21	C2327J5003AHF phase balance	Spec typ 4 deg / max 8 deg vs plot ~2.5-3.2 deg over 2.4-2.6 GHz; broadband plot shows >= 20 deg above 4.4 GHz (part not usable there).
I22	Two datasheet ages	TA1442A approval sheet is from 2011 (rev. 1): check the part is still in production before design-in.
I16	AN #18 scope	AN #18 covers 315/434/868 MHz RKE receivers (TDA520x, ATA572x). Only its matching method applies here; its component values do not.

Table - Inconsistencies.

Part	Missing parameter	Treatment
TQP3M9037	Noise parameters below 1.5 GHz (L2/L5 band)	Estimated (A3)
TQP3M9037	S-parameters at 3.3 V (only 4 typical points)	Not needed (5 V chosen)
TQP3M9037	Fine-step S-parameters (table step 200 MHz)	Interpolated (A6)
TQP3M9037	P1dB / OIP3 at 1.18-1.6 GHz	Interpolated (A7)
HC125A	Absolute S-parameter phase, reflection phase, data outside 1150-1630 MHz	Assumed (A8, A11)
B7504	Phase / group delay outside the GLONASS band, in-band power rating at 1559-1606 MHz, IIP3	Min-phase estimate; not rated
B8889	Group delay, phase, IIP3	Min-phase estimate
DXP.02.A	Phase, group delay, out-of-band data above 2.2 GHz, power-handling per port	Min-phase estimate; held flat

Part	Missing parameter	Treatment
All	Temperature drift of the SAW centre frequency (typ. -30...-40 ppm/K for SAW)	Not modelled
TA1442A	Phase / group delay, power handling per frequency, IIP3	Min-phase estimate; 10 dBm max input only
C2327J5003AHF	Reflection at ports other than the input, absolute phase	Assumed (A23)
All	Package/PCB parasitics for the layout	Not modelled (A17)

Table - Missing parameters.

14 Limitations

- SAW and diplexer Touchstone files are estimates from digitized plots: magnitude accuracy +/-0.3 dB in band and +/-2 dB in the stop band; phase and group delay are minimum-phase estimates (real SAW delay can be larger).
- HC125A phase is assumed; only magnitudes and the coupled/through phase difference are datasheet values. The HC125A is held flat outside 1150-1630 MHz.
- The antenna is an ideal RHCP source; patch impedance, feed coupling, axial ratio and pattern are not modelled.
- LNA data are 25 C typical at 5 V with 200 MHz steps; no temperature, supply, part-to-part or aging spread.
- No PCB interconnect, via, ground or coupling effects; no layout-induced feedback around the 40 dB of gain.
- Noise below 1.5 GHz and large-signal values at L-band are interpolated/estimated.
- Interference analysis is small-signal (no compression, no intermodulation products, no pulsed DME statistics).
- AN #18 is an RKE (315/434/868 MHz) application note; its method is used, its component values are not.

15 Recommendations

- Add a pre-LNA filter or at least a DME/L-band notch: the L2/L5 branch needs > 30 dB at 960-1150 MHz; e.g. move one B8889 in front of LNA 1 (costs ~0.9 dB NF) or use a ceramic/cavity L5 pre-filter. For L1, a pre-filter with steeper upper skirt (Iridium) such as B39162B2611P810 (0.9 dB, pre-LNA) from the SDST.
- Reduce gain to what the receiver needs (typ. 25-30 dB at the connector): drop to one LNA per branch, or keep two and run LNA 2 at 3.3 V (-1.5 dB gain, -4 dBm OP1dB, 44 mA) or add a pad - this also eases the SAW 2 power rating.
- Improve the RHCP-port match: use the Qorvo optimized-return-loss input network (3.3 pF series + 3.9 nH shunt, p.9) or a two-step AN #18 style match; the hybrid then also sees a better load, improving combining/axial ratio.
- Choose the second B8889 position or replace B8889 by B2642/B2632 if GLONASS G2 is required (B8889 stops at ~1245 MHz).
- Get measured data: RF360 provides measured .s2p for B7504/B8889; Taoglas for HC125A (.s4p) and DXP.02.A (.s3p); Qorvo for the TQP3M9037 fine-step .s2p and low-band noise parameters. Drop them into Touchstone/ and rerun Scripts/build_all.py - every result regenerates.
- Layout: keep 40 dB of gain stable - separate input and output grounds of each LNA with via fences, keep the SAWs between the LNAs, follow the Qorvo via pattern (p.15) and HC125A footprint (p.11-13).
- Next simulation step: import the real patch S-parameters (HFSS) in place of PatchCP_ideal, and add MSL models for the interconnects in AEDT Circuit.

16 Deliverables

Folder	Content
Touchstone/	TQP3M9037_5V.s2p (extracted, with noise), TQP3M9037_stage_5V_with_bias.s2p, HC125A.s4p, B7504.s2p, B8889_matched.s2p, DXP02A.s3p, PatchCP_ideal_RHCP_feed.s4p (estimated); SIM_*.s2p/.s3p (simulated results)
Scripts/	ds_data.py, models.py, simulate.py, build_extracted.py, make_schematics.py, aedt_circuit.py (PyAEDT), build_circuitrf.py, build_qucs.py, crosscheck_heavi.py, make_report.py, build_all.py
ANSYS_Project/	GNSS_L1_L2L5_FrontEnd.aedt (when AEDT runs), AEDT_System_3port.s3p, aedt_run.log
circuitRF/	gnss_frontend.cnl netlist and its circuitRF S-parameter result
QucsS/	gnss_frontend.net Qucsator-RF netlist and its S-parameter result
Report/	this report (PDF and DOCX), error_log.md
Images/	schematics (PNG + SVG), connection diagram, all result plots
Extracted_Data/	36 CSV tables, Extracted_Parameters.xlsx, assumptions_log.md, results_*.csv, cross-check JSON

Table - Deliverables.

Appendix A - Error log

#	Step	Error	Resolution / status
E1	Topology	Chain as first written (HC125A - LNA - filter - LNA - filter - DXP) cannot pass both bands: B7504 (L1) and B8889 (L2/L5) have disjoint pass-bands	Asked the user; two-branch topology with the DXP as combiner confirmed
E2	Tool install	`pip install ansys-aedt-core` not found on the package index	Installed `pyaedt` 1.9.0 (provides `ansys.aedt.core`)
E3	Tool install	`uvx` install of the PyAEDT MCP server from GitHub blocked by the permission classifier	Not installed; user asked to run it themselves if wanted
E4	Tool install	heavi 0.7.9 fails to import on Python 3.12 (`mutable default UnassignedType` dataclass error) and downgraded the global numpy	heavi moved to the repo `.venv`; one-line local fix (`__hash__ = object.__hash__` in `heavi/node.py`); global numpy restored to 2.5.4
E5	Tool install	circuitRF portable zip: `unzip` warns about backslash paths	Extracted with PowerShell `Expand-Archive`; MCP server registered locally (`claude mcp add circuitrf`)
E6	HC125A model	Table magnitudes with arbitrary reflection/isolation phases gave a non-passive 4-port (largest singular value 1.05)	Phases chosen to minimise the largest singular value (1.013); residual scaling <= 0.11 dB (assumption A10)
E7	circuitRF	`Token overflow: got 17 tokens, expected 9 for 2-port` when reading MHz / DB Touchstone files	All Touchstone files rewritten as GHz / RI; circuitRF then reads them
E8	heavi cross-check	First comparison differed by up to 1.4 dB	Cause: linear vs cubic interpolation of the 200 MHz LNA table and missing parasitics. With identical models the difference is < 0.01 dB; the interpolation sensitivity is reported as model uncertainty (A6)
E9	ANSYS AEDT	PyAEDT 1.9 + AEDT 2026 R1: `Failed to start new AEDT gRPC session` (transport mode WNUA) from both Bash and PowerShell; `ansysedt -grpcsrv` process starts but does not open the port within 10 min	Not solved in this session. Run `python Scripts/aedt_circuit.py --gui` on the workstation (an interactive AEDT can answer licence/first-run prompts), or `--version 2024.1` for the 2024 R1 install. Two idle `ansysedt.exe` processes from these attempts (PIDs 9720, 10988) could not be closed from the session (permission denied) - close them in Task Manager

Table - Errors met while building this project and how they were resolved.